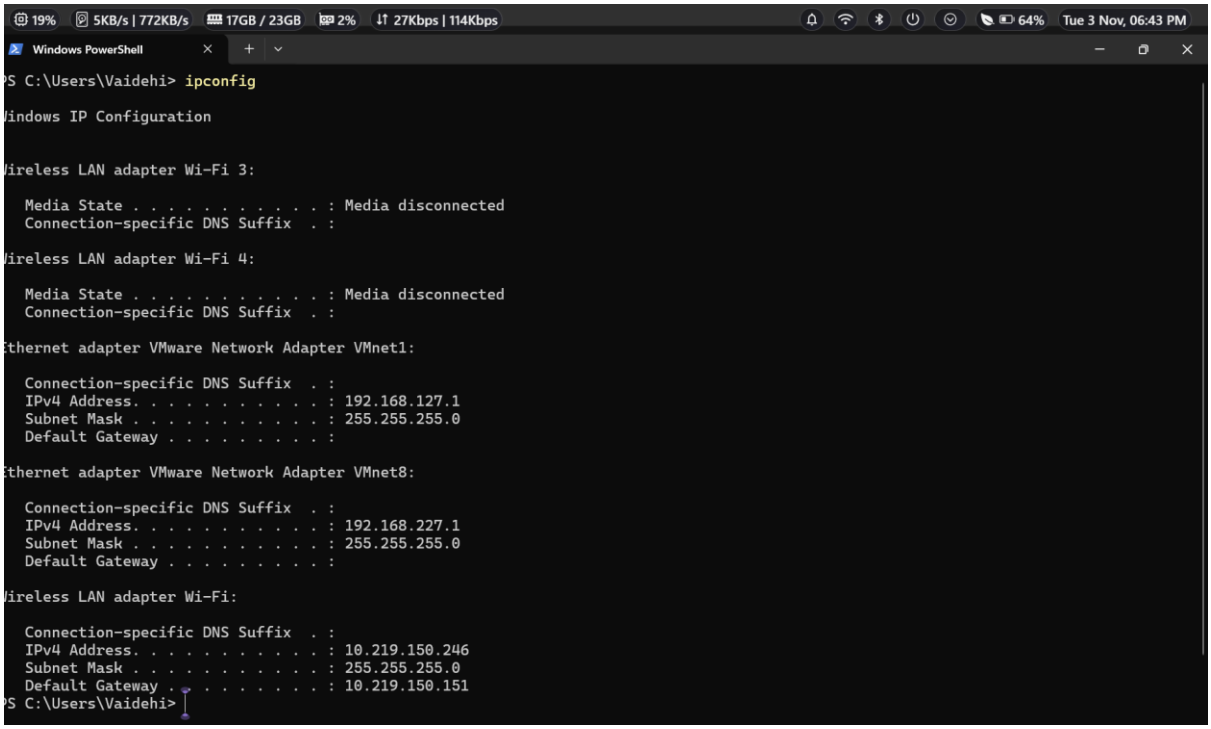
IP Addressing

Task – 1 : Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.

Solution :



Field	Value
IPv4 Address	10.219.150.246
Subnet Mask	255.255.255.0
Default Gateway	10.219.150.151

It’s a Private IPv4 address.

Task – 2 : Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89.
Hint : Check the standard private IP address ranges for IPv4.

Solution : IPv4 private addresses are reserved for use within local networks. There are three private IPv4 address ranges:

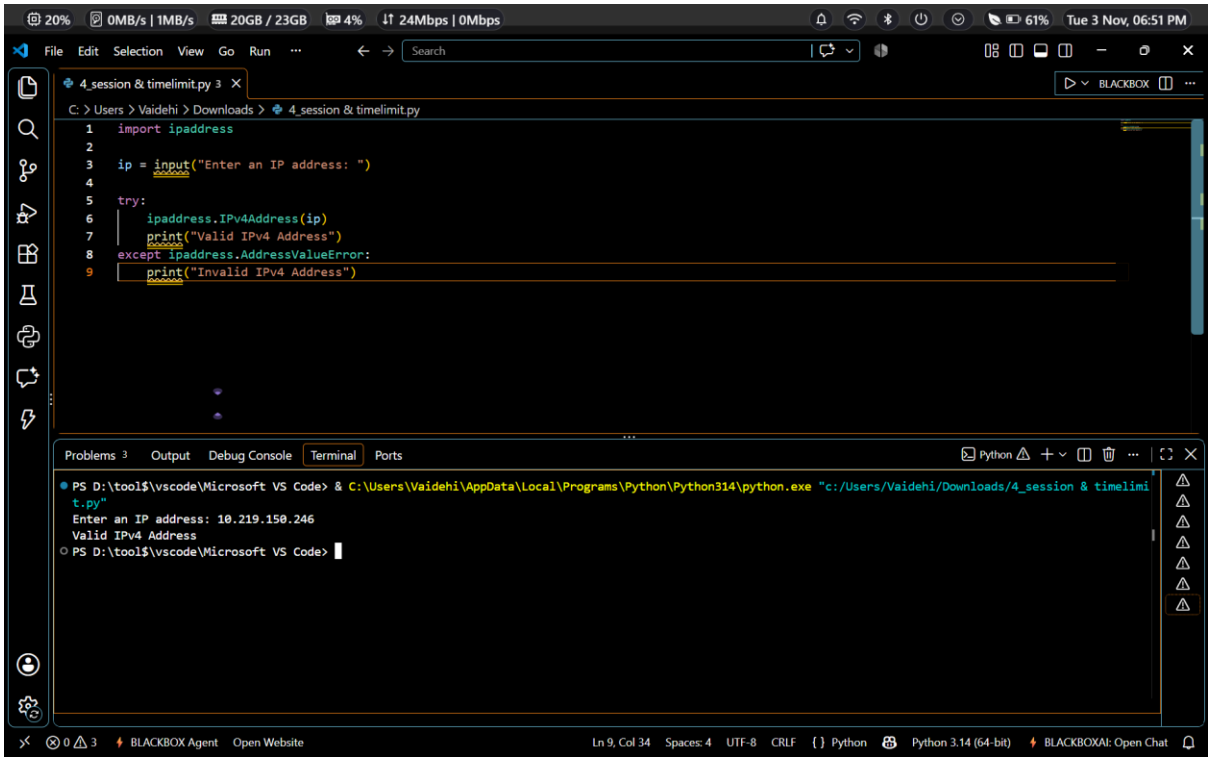
1. 10.0.0.0 to 10.255.255.255
2. 172.16.0.0 to 172.31.255.255
3. 192.168.0.0 to 192.168.255.255

Classification of Given IP Addresses:

IP Address	**Type**	**Explanation**
192.168.1.10	Private	Belongs to the 192.168.0.0/16 range.
10.0.0.5	Private	Belongs to the 10.0.0.0/8 range.
172.16.5.20	Private	Belongs to the 172.16.0.0/12 range.
8.8.8.8	Public	Does not belong to any private IP range.
123.45.67.89	Public	Does not belong to any private IP range.

Task – 3 : Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not. Constraint : Do not use any third-party libraries for validation.

Solution :



Explanation:

1. The program asks the user to enter an IP address.
2. The IPv4Address() function checks whether the entered address follows IPv4 format.
3. If the address is valid, it displays "Valid IPv4 Address".
4. If the address is invalid, it displays "Invalid IPv4 Address".

Task – 4 : Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.

Solution : Instagram and WhatsApp use IP addresses as one of the methods to monitor login activity and identify potentially suspicious access to user accounts.

Example Scenario:

Suppose I log in to my Instagram account from Ahmedabad, India.

- First Login: Instagram records the login IP address associated with my internet connection in Ahmedabad.
- Second Login: Someone attempts to access my account using a different IP address associated with a network in another country, such as the USA.
- Suspicious Activity Detection: Instagram may compare the login locations, device information, and login times.
- Security Alert: If the activity appears unusual, Instagram may request additional verification or send a security notification.
- Account Protection: The user can review the login activity, change their password, and enable two-factor authentication if necessary.

How IP Addresses Help in Security:

1. Location Detection: IP addresses can provide an approximate geographical location of a login.
2. Unusual Login Detection: Logins from unexpected locations or networks may trigger security checks.
3. Account Protection: Suspicious login attempts can lead to additional authentication requirements.
4. Login Monitoring: Users can review devices and sessions associated with their accounts.